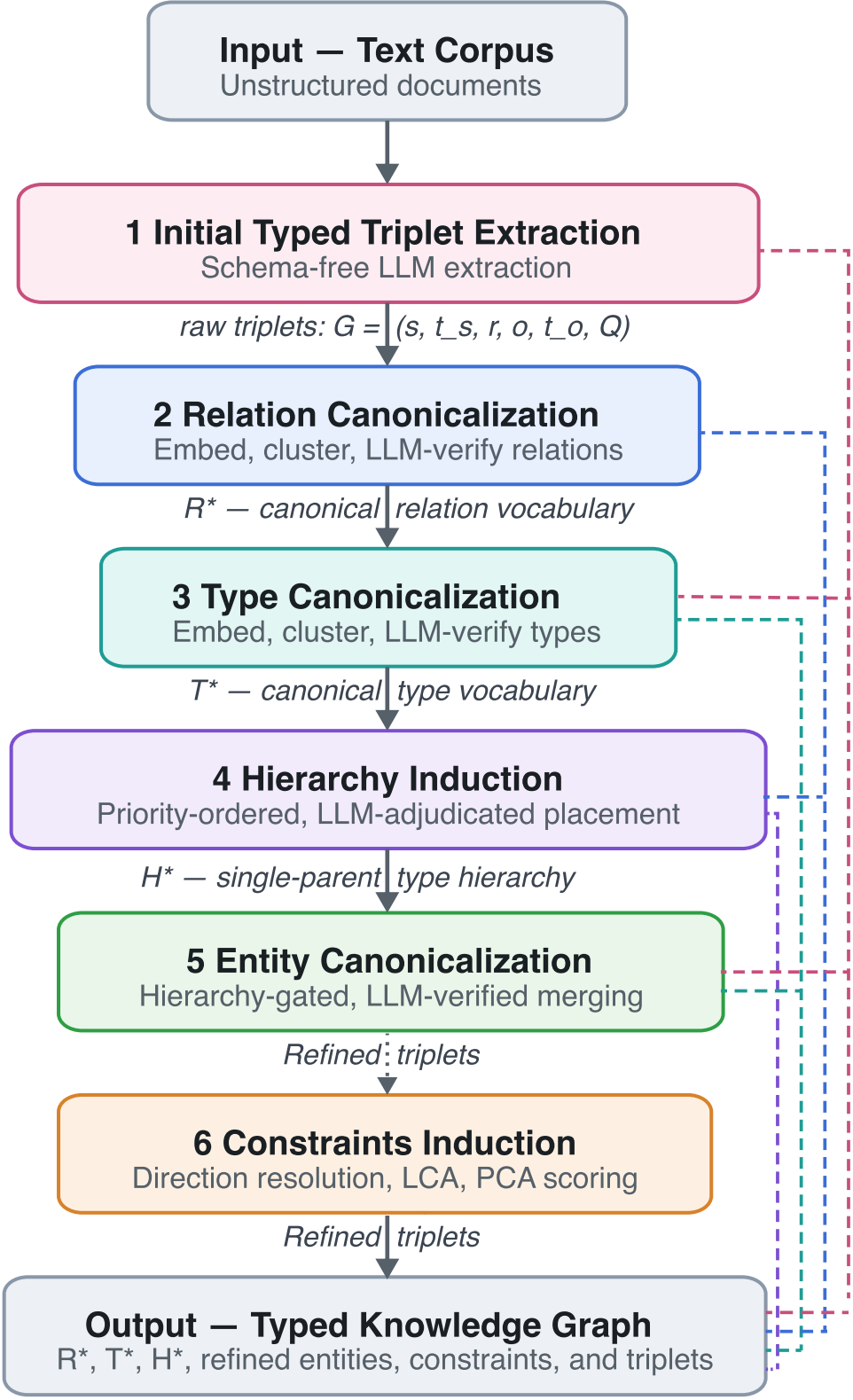

Input — Text Corpus
Unstructured documents
1 Initial Typed Triplet Extraction
Schema-free LLM extraction
raw triplets: G = (s, t_s, r, o, t_o, Q)
2 Relation Canonicalization
Embed, cluster, LLM-verify relations
R* — canonical relation vocabulary
3 Type Canonicalization
Embed, cluster, LLM-verify types
T* — canonical type vocabulary
4 Hierarchy Induction
Priority-ordered, LLM-adjudicated placement
H* — single-parent type hierarchy
5 Entity Canonicalization
Hierarchy-gated, LLM-verified merging
Refined triplets
6 Constraints Induction
Direction resolution, LCA, PCA scoring
Refined triplets
Output — Typed Knowledge Graph
R*, T*, H*, refined entities, constraints, and triplets